

MUHAMMAD GEMILANG RAMADHAN

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Summary

Front-End Engineer (JS/TS, React, Next.js, Tailwind) delivering performant, accessible web apps. Built multi-tenant e-commerce UI with checkout, payments (Midtrans API for QRIS), maps, and POS sync; cut perceived load time via HTTP/query caching, code-splitting, and lazy loading. Comfortable with Node/Express, REST, WebSocket.

Skills

- **Languages:** HTML, CSS, JavaScript (ES6+), TypeScript, Python, PHP, SQL (MySQL)
- **Front-End:** React, Redux, Tailwind CSS, Bootstrap, Vite
- **Back-End:** Node.js, Express, REST APIs, JWT, WebSocket
- **ML/CV/RL:** TensorFlow, PyTorch, OpenCV, NumPy, Pandas, stable-baselines3
- **Tools:** Git, GitHub, Jira, Notion
- **Languages (human):** Indonesian (native), English (fluent)

Projects

Kulkas Babe – multi-tenant e-commerce (Frontend-side) | June 2025 – August 2025

- Sole front-end in a 2-person team to build responsive, dynamic shopping UI/UX with Laravel Blade, Alpine.js, Tailwind, vanilla JS.
- Designed & optimized cart, checkout, and payment flows. Integrated Midtrans API (QRIS), Google Maps API (distance/fees), and Olsera POS (catalog sync).
- Implemented caching & performance tactics (HTTP/query caching, asset minimization, lazy loading) to cut load times and stabilize UX.
- Partnered with back-end to enforce secure, maintainable components (CSRF, validation, auth guards), versioned patterns, and clean code reviews.

JejakBatik — ML Pattern Recognition (Bangkit Capstone, team of 5) | December 2024

- Architected the end-to-end ML pipeline (data prep, augmentation, training, evaluation) and designed an EfficientNetB0 classifier, achieving 85% pattern-recognition accuracy.
- Implemented real-time image processing & classification with TensorFlow (and TensorFlow JS for mobile preparation) + OpenCV, integrating the model with the Android app for on-device inference.
- Tuned preprocessing and caching to keep inference responsive.
- **Stack:** Python, TensorFlow (EfficientNetB0), OpenCV.

Autonomous Drone Navigation — DDPG on AirSim | 2025

- Designed and trained DDPG (actor-critic-based algorithm) with stable-baselines3 to control a quadrotor in cluttered environments on AirSim.
- Built the RL pipeline: environment integration, reward shaping (progress-to-goal, collision penalties), and an evaluation harness (N-episode rollouts).
- Result: ~80% episode success rate with high average reward return. Policy consistently avoided obstacles and reached targets efficiently.
- **Stack:** Python, stable-baselines3 (DDPG), PyTorch, AirSim, NumPy.

Web Developer & Designer — Anargya ITS EV Team (Student Org) | January 2024 – February 2024

- Shipped the team's public website; translated Figma designs into responsive, accessible pages using Bootstrap.
- Built a lightweight brand system (colors, type scale, spacing) to keep web and Instagram assets consistent.
- Produced social assets (feed/story carousels, sponsor spotlights) that supported announcements and partner visibility

Education and Certification

- **Bachelor Program (expected to graduate in late 2025)**
Bachelor's degree – B.S., Electrical Engineering. Institut Teknologi Sepuluh Nopember (ITS).
- **Bangkit Machine Learning Path 2024 Graduates | 2024**

I graduated as a Machine Learning developer and got my ML and TensorFlow certification. Created a lot of machine learning projects such as mastering the unsupervised and supervised learning, OpenCV for image segmentation, and pattern recognition.

- **Udemy - 100 Days of Python | 2023**

Course: A bunch of series of Python programming courses, including projects, basic functions, development, etc.

- **MySkill - Ms. Excel in MySkill. | 2023**

Course: Mastering basic skills of Excel including pivot table, power pivot, and data visualization.

- **SQL for Data Analysis in MySkill. | 2023**

Course: Learning the fundamentals of SQL that are required for data analysis, manipulating table, etc.